Extreme Value Analysis of the USD/EUR Exchange Rate

An empirical approach to modeling heavy tails

Project Report — *Extreme Value Analysis* Course

Data: Daily USD/EUR exchange price, January 2004 – December 2023

Members:

Alexandre Picart

Badiallo Kantako

Lucien Mirlicourtois

Master 2 Econometrics & Statistics

Group 4

Professor:

Mr. Abdelaati Daouia

"What is the extreme value behaviour of the USD/Euro exchange rate?"

Contents

Introduction **2**

1 Data Exploration and Preparation **2**

1.1 Data presentation and Quality Control . . . 2

1.2 Price Series Analysis and Non-Stationarity . . . 2

1.3 Transformation into Log-Returns and stationarity checking . . . 3

1.4 Checking of normality distribution . . . 4

2 Extreme Value Analysis — Modeling **5**

2.1 Tail Index Estimation: A Pseudo-Likelihood Method . . . 5

2.1.1 Graphical Approach: Hill and Moment estimators . . . 5

2.1.2 Optimal k Selection via the dAMSE Algorithm . . . 6

2.1.3 Validation of the choice of k — Pareto QQ-plot . . . 6

2.2 Block Maxima (BM) Method . . . 7

2.2.1 Choice of Block Size . . . 7

2.2.2 GEV Parameter Estimation . . . 7

2.2.3 GEV Residuals Diagnostics . . . 8

2.3 Peaks Over Threshold (POT) Method . . . 9

2.3.1 Selection of Threshold u . . . 9

2.3.2 GPD Parameter Estimation . . . 10

2.3.3 Diagnostics and GPD Model Validation . . . 10

2.3.4 Return Levels and Bootstrap Confidence Intervals . . . 11

2.3.5 Extreme Risk Metrics: Value at Risk (VaR) and Expected Shortfall (ES) . . . 11

3 Handling Serial Dependence **12**

3.1 Data aggregation approaches and Validation of Independence . . . 12

3.2 Statistical Inference: Tail Heaviness and Symmetry . . . 13

Conclusion **14**

References **15**

A Theoretical Appendix: Foundations of Extreme Value Theory **16**

A.1 Fisher-Tippett-Gnedenko Theorem (Limit of Maxima) . . . 16

A.2 Pickands-Balkema-de Haan Theorem (Limit of Exceedances) . . . 16

A.3 Relevance for Financial Risk Management . . . 16

B Supplementary Material Appendix **17**

Introduction

Foreign exchange markets constitute the most liquid segment of global finance, with a daily volume exceeding $7000 billion Bank for International Settlements 2022. The EUR/USD pair, the most traded in the world, accounts for nearly 22.7% of all transactions. However, its extreme fluctuations whether triggered by the 2008 financial crisis, the COVID-19 pandemic in 2020, or the 2022 energy shock can generate considerable losses for financial institutions, multinational corporations, and asset managers.

Classical approaches implicitly assume that returns follow a Gaussian distribution; however, foreign exchange markets are regularly subjected to sudden shocks (monetary policy decisions, geopolitical events) that produce observations far beyond theoretical predictions. Ignoring the thickness of these distribution tails amounts to structurally underestimating risk, which has direct consequences on the sizing of hedges and capital requirements. To address this flaw, **Extreme Value Theory** (EVT) offers a rigorous mathematical framework. Based on asymptotic convergence theorems, it allows for the robust modeling of tail distribution behavior.

Therefore, this report aims to answer the following question: ***What is the extreme behavior of the USD/EUR exchange rate?***

To carry out this analysis, we will proceed as follows: first, we conduct an exploratory analysis and data engineering to prepare the dataset. Second, we estimate the tail index for both extremities (left and right tails) using various estimators. Third, we implement solutions to correct serial dependence. Finally, we perform formal inference and symmetry tests to characterize the exchange rate behavior, before concluding with the summary, bibliography and appendices.

1 Data Exploration and Preparation

1.1 Data presentation and Quality Control

To conduct this analysis, we rely on a historical dataset covering the period from **January 2004 to December 2023**, totaling 5217 daily observations sourced from the `USDEUR.csv` file. The quotation studied here technically corresponds to the USD/EUR rate: it represents **the value of one US dollar expressed in euros** (1 USD $= x$ EUR).

The raw data includes six variables for each trading day: the session date (`Date`), the opening price (`Open`), the highest (`High`) and lowest (`Low`) levels reached, the raw closing (`Close`), and the adjusted closing price (`Adj.Close`).

After converting the variables to the appropriate numeric format, the initial exploration of our raw database revealed the presence of 29 missing values, distributed across all the rating columns (with the exception of the variable `Date`). These observations were completely removed. Furthermore, a rigorous check confirmed the absence of duplicate dates: no problematic or duplicate dates were detected.

Following this cleaning, the database is perfectly clean. For the remainder of our analysis of the extreme risks of the USD/EUR exchange rate, we will use the daily closing price to start our analysis.

1.2 Price Series Analysis and Non-Stationarity

When working with time series, it is essential to verify the stationarity of the data before any modeling. Visual observation of the raw closing price series (see Figure 1.1), along with the ACF and PACF, shows a very slow decay characteristic of a non-stationary process. This graphical observation is confirmed

by stationarity tests (notably the Augmented Dickey-Fuller test), which do not allow us to reject the hypothesis of a unit root.

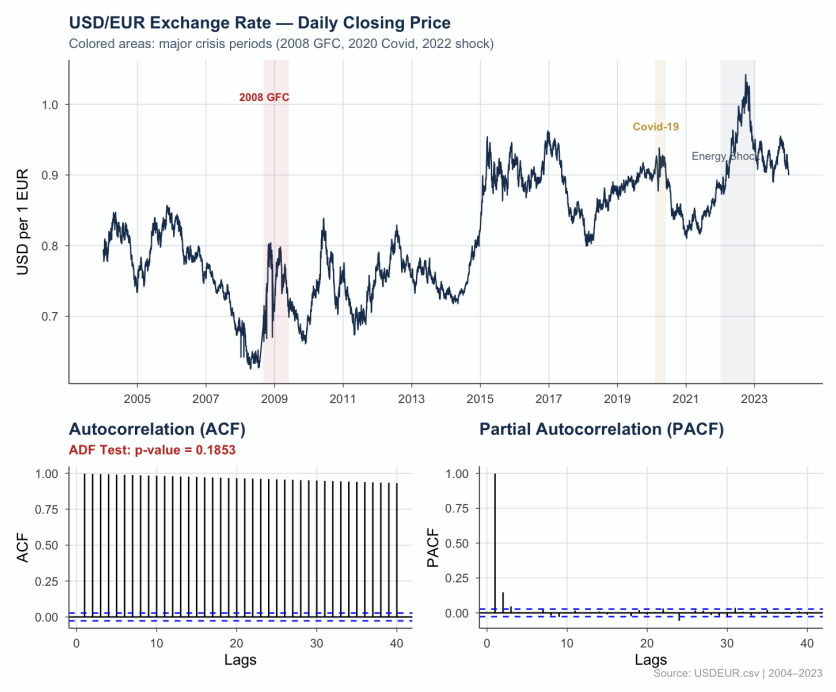


Figure 1.1. Time series and stationarity check of the raw closing price

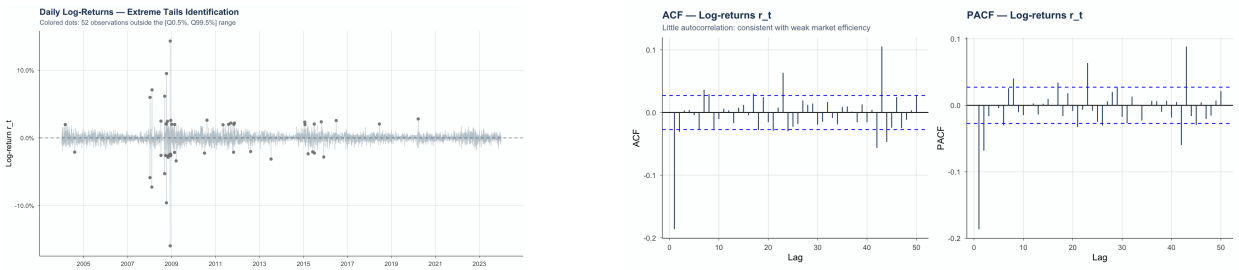
The closing price of the USD/EUR exchange rate is a series that evolves over time with a changing trend and variance. Between 2004 and 2023, it went through very different regimes. Such a series is said to be non-stationary: its distribution changes depending on the period considered. However, the entire theory of extreme values, the Fisher-Tippett and Pickands theorems, rests on a fundamental assumption: the observations must come from a distribution that is stable over time. Applying Hill, GEV, or GPD directly to this series would yield estimators without a theoretical basis.

1.3 Transformation into Log-Returns and stationarity checking

To overcome this problem and obtain a usable series, we apply a logarithmic transformation to extract continuous returns. The log-return r_t is calculated using the first difference of the logarithm of prices:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) = log(P_t) - log(P_{t-1}) \tag{1.1}$$

The transformation of prices into log-returns is a fundamental prerequisite for our analysis. Primarily, it stationarizes the data, a mathematical necessity for EVT estimators that is strongly validated by our unit root tests. Furthermore, log-returns ensure temporal homogeneity by capturing scale-independent relative variations, thereby making historical extreme events (such as those in 2008 and 2020) directly comparable. Finally, their linear additivity over time is a crucial property for accurately extrapolating multi-year return levels within the Peaks Over Threshold (POT) framework.



(a) Log-returns evolution **(b)** ACF/PACF correlograms

Figure 1.2. Exploratory analysis of USD/EUR log-returns

A visual examination of the log-return series (see Figure 1.2a) reveals generally stable fluctuations around a zero mean, providing initial evidence of stationarity. This dynamic is complemented by a visual analysis of the autocorrelation function (ACF), whose rapid decay to zero is characteristic of a stationary process. Statistical tests now strongly confirm that we reject the hypothesis of non-stationarity: our series has become stationary (see Table B.1).

However, a more detailed analysis of correlograms (ACF/PACF) reveal significant lags confirmed by formal tests (Appendix B.2), indicating an autocorrelation of returns. Furthermore, the study of squared returns (r_t^2) highlights persistent heteroscedasticity (**ARCH effects** or volatility clustering).

Thus, although stationary, the series does not behave like strictly independent and identically distributed (i.i.d.). This dependence does not bias the point estimates, but it invalidates the standard asymptotic inference (confidence intervals and tests become too optimistic). Being aware of this limitation, we employ alternative data aggregation strategies in the subsequent sections to mitigate this dependence and ensure the reliability of our conclusions.

1.4 Checking of normality distribution

To justify the use of Extreme Value Theory (EVT), it is necessary to show that the empirical distribution of USD/EUR log returns deviates significantly from a normal distribution. The log return histogram (Figure 1.3) exhibits tails that are considerably thicker than those of a normal distribution, as well as a very narrow central peak, and the QQ plot visually confirms the thickness of the tails, particularly at the extremes, where the points diverge from the diagonal. This thickness is quantified by an excess of kurtosis of 98.02 (compared to 0 for the normal distribution) and a slightly negative skew of 0.585. Formal normality tests (Jarque-Bera, Kolmogorov-Smirnov, Anderson-Darling) all reject the null hypothesis with p-values close to zero (see Table B.3).

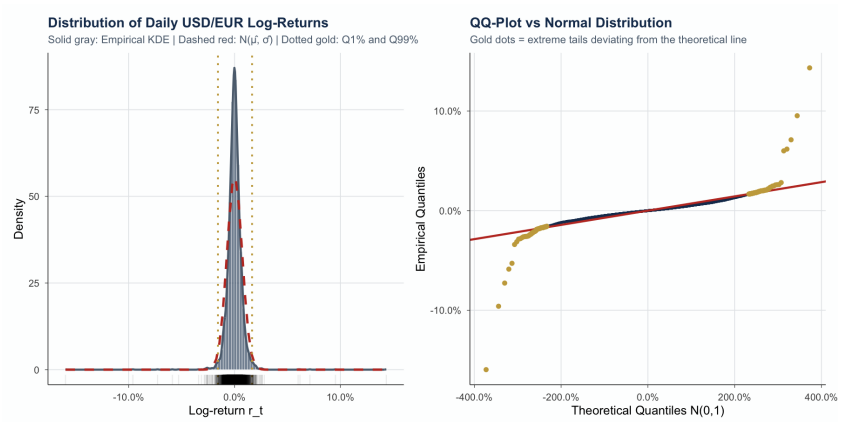


Figure 1.3. Empirical distribution of USD/EUR log-returns compared to a standard Normal distribution

The distribution is therefore not Gaussian. This finding justifies the use of Extreme Value Theory (EVT) to model the extreme behavior of the exchange rate. The precise diagnosis of asymptotic heaviness (Fréchet vs. Gumbel domains) will be established using the Hill estimator and GPD/GEV models.

2 Extreme Value Analysis — Modeling

2.1 Tail Index Estimation: A Pseudo-Likelihood Method

To rigorously analyze the extreme risk behavior of the USD/EUR exchange rate, it is essential to study both ends of the distribution. We partition our sample to evaluate the right tail (extreme daily gains, $x = r_t > 0$) and the left tail (extreme daily losses, transformed to positive values $x = -r_t > 0$). Our primary objective is to estimate the Extreme Value Index (EVI), denoted γ, which characterizes the heaviness of these tails.

2.1.1 Graphical Approach: Hill and Moment estimators

To quantify this parameter, we primarily rely on the Hill estimator. However, to ensure the robustness of our results, we plot it alongside the Bias-Reduced Hill estimator (RB-Hill) and the Moment estimator (Dekkers-Einmahl-de Haan). As observed in Figure 2.1, plotting these estimators against the number of upper order statistics k illustrates the classic bias-variance trade-off. For very small values of k, the estimators exhibit high variance and oscillate widely. Conversely, as k becomes too large, a severe downward bias appears: the inclusion of central observations dilutes the extreme nature of the sample, causing the estimators to artificially underestimate the true thickness of the tail.

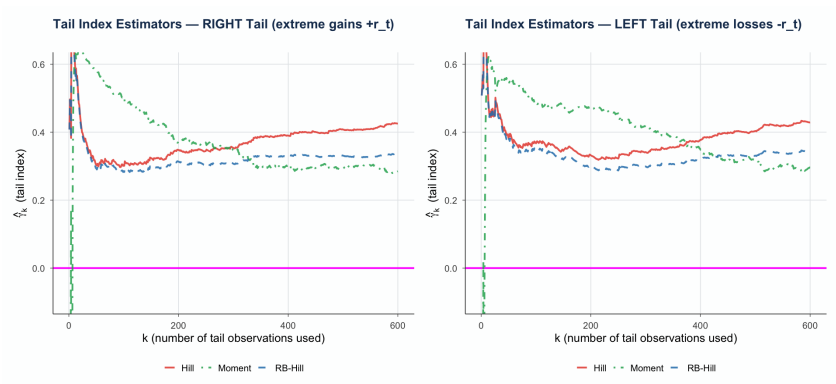


Figure 2.1. Tail index estimator by Hill, RB-Hill and MM

To formally confirm the heavy-tailed nature of our data, we plot these estimators alongside a magenta horizontal line representing $\gamma = 0$. For both tails, all estimators strictly and consistently evolve above this line. Most notably, the positive trajectory of the Moment estimator which does not assume a heavy tail a priori—formally confirms that the data belongs to the Fréchet domain of attraction ($\gamma > 0$). Furthermore, overcoming the initial volatility, a clear stability region (a "plateau") emerges for k ranging approximately between 120 and 200. In this region, the RB-Hill estimator effectively smooths the drift caused by central observations, yielding a estimate of γ between 0.2 and 0.4.

2.1.2 Optimal k Selection via the dAMSE Algorithm

While the visual identification of the stability plateau between $k = 120$ and $k = 200$ is reassuring, graphical selection remains inherently subjective. To rigorously determine the exact optimal number of order statistics k^*, we apply an automated algorithmic procedure.

Specifically, we use the double-bootstrap methodology to minimize the Asymptotic Mean Squared Error (dAMSE) of the estimator. This approach automatically pinpoints the threshold k^* that perfectly balances variance and bias within our visual plateau.

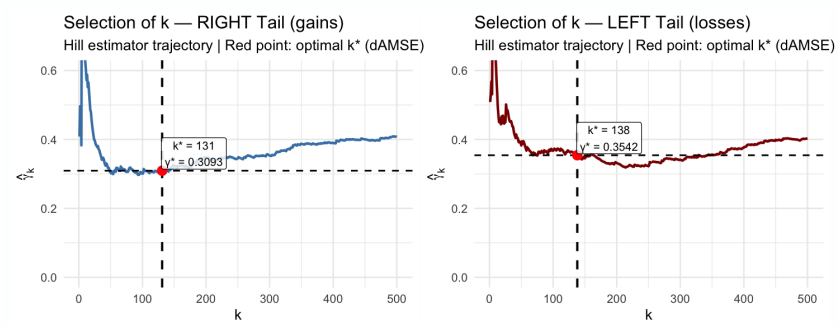


Figure 2.2. Optimal k selection

As summarized in Table B.5, the optimal tail indices are $\gamma^* \approx 0.31$ for the extreme gains and $\gamma^* \approx 0.35$ for the extreme losses. Both values are strictly positive, showing that both ends of the USD/EUR distribution exhibit heavy tails belonging to the Fréchet domain of attraction.

2.1.3 Validation of the choice of k — Pareto QQ-plot

To assess the quality of our tail estimation, we perform a diagnostic check using the QQ-plots presented in Figure 2.3. These plots compare the empirical quantiles of the exceedances against the theoretical quantiles of a heavy-tailed distribution (generalized Pareto) based on the dAMSE-optimal γ^*.

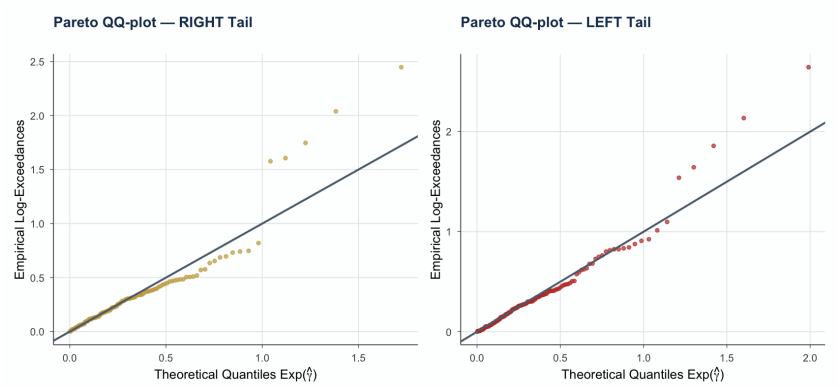


Figure 2.3. Diagnostics of dAMSE estimator

For both tails, the observations are closely aligned with the diagonal reference line, indicating that the heavy-tailed model provides an excellent fit for the extremes of the USD/EUR exchange rate. While slight departures are visible at the very far ends of the distribution, these are expected given the reduced sample size in the tail region. Overall, this diagnostic confirms that our choice of k^* and the subsequent tail index estimates are statistically reliable for quantifying extreme risks.

2.2 Block Maxima (BM) Method

To complement the tail index analysis, we now apply the Block Maxima (BM) approach. According to the Fisher–Tippett–Gnedenko theorem (see A.1) , if maxima are taken over sufficiently large non-overlapping blocks, their distribution can be approximated by a Generalized Extreme Value (GEV) law. In practice, the main modeling choice is therefore the block size, which must balance two constraints: blocks should be large enough for the GEV approximation to be meaningful, while still leaving a sufficient number of maxima for reliable estimation.

2.2.1 Choice of Block Size

In our case, we divide the series of daily log-returns into monthly blocks and retain, for each month, the largest gain and the largest loss in absolute value. This choice appears natural for financial data observed at daily frequency. Each block contains around 21 trading days, which is large enough to summarize local extreme behavior, while the full period from January 2004 to December 2023 still provides 240 monthly blocks, ensuring a sufficient number of maxima for estimation. This monthly aggregation therefore offers a reasonable compromise between approximation bias and estimation variance.

The corresponding series of monthly maxima is displayed in Figure 2.4. It reveals pronounced spikes during major crisis episodes, especially around the 2008 global financial crisis, the Covid shock in 2020, and, to a lesser extent, the 2022 turmoil. This already suggests that the most extreme exchange rate movements are concentrated in periods of financial stress.

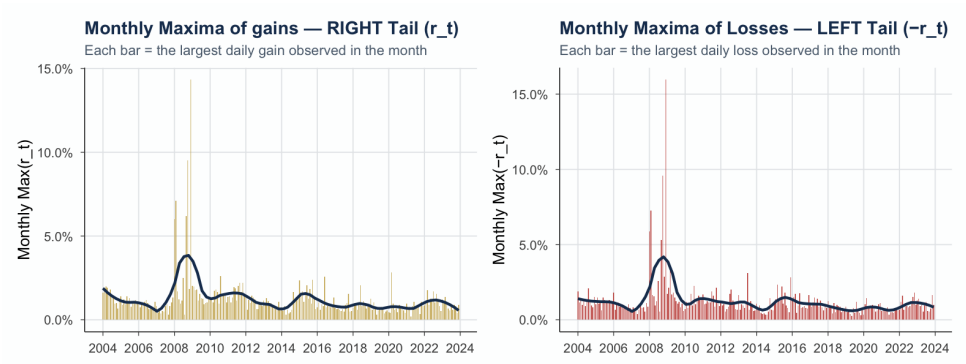


Figure 2.4. Monthly block maxima of USD/EUR daily log-returns for the gain and loss tails.

2.2.2 GEV Parameter Estimation

We then fit a Generalized Extreme Value distribution to the monthly block maxima by maximum likelihood. In the Block Maxima framework, the GEV distribution is characterized by three parameters: a location parameter μ, a scale parameter σ, and a shape parameter ξ. Among them, the shape parameter is the most important, since it determines the nature of the tail behavior. A positive value of ξ indicates a Fréchet-type tail, associated with heavy-tailed behavior, whereas $\xi = 0$ corresponds to the Gumbel case and $\xi < 0$ to the Weibull case.

The estimation results are reported in Table 2.1. For the right tail (extreme gains), we obtain $\hat{\xi} = 0.2576$, while for the left tail (extreme losses), the estimated shape parameter is $\hat{\xi} = 0.2799$. In both cases, the point estimates for the shape parameter are strictly positive, which is **consistent with** the Fréchet domain of attraction.

Table 2.1. GEV Parameter Estimates for Monthly Block Maxima ($n = 240$)

Queue	Parameter	Estimate	SE	$\mathbf{CI}^{inf}_{95\%}$	$\mathbf{CI}^{sup}_{95\%}$
Right (Gains)	μ (loc)	0.008115	0.000279	0.007568	0.008663
Right (Gains)	σ (scale)	0.003924	0.000178	0.003576	0.004272
Right (Gains)	ξ (shape)	**0.257613**	0.047502	0.164509	0.350717
Left (Losses)	μ (loc)	0.008128	0.000271	0.007597	0.008659
Left (Losses)	σ (scale)	0.003765	0.000172	0.003429	0.004101
Left (Losses)	ξ (shape)	**0.279882**	0.050590	0.180726	0.379039

These results, obtained through the Block Maxima approach, provide further evidence of heavy-tailed behavior in the USD/EUR exchange rate log-returns. The proximity of the two shape parameters suggests a relatively symmetric tail structure; however , this observation remains purely descriptive at this stage. A formal assessment of these parameters' significance and a rigorous symmetry test are deferred to **Section 3**, where robust methods will be used to account for the series' underlying dependence and ensure the validity of our conclusions.

2.2.3 GEV Residuals Diagnostics

To assess the adequacy of the GEV specification, we examine the standard diagnostic plots associated with the fitted models, namely the probability plot, quantile plot, density plot, and return level plot. These diagnostics are reported in Figure 2.5 for the gain tail and Figure 2.6 for the loss tail.

Overall, the diagnostics suggest that the GEV approximation provides a satisfactory description of the monthly block maxima. In particular, the probability plots are very close to the diagonal for both tails, which indicates that the fitted distributions reproduce the empirical probabilities rather well. The density plots also show that the fitted GEV model captures the main shape of the empirical distribution of maxima.

Some departures nevertheless remain visible in the quantile plots and, more clearly, in the return level plots for the most extreme observations. These deviations are concentrated in the upper part of the sample, where the empirical maxima become more dispersed than the fitted theoretical values. Such differences are not surprising, since the Block Maxima method relies on a limited number of maxima, so the largest observations are necessarily estimated with substantial uncertainty.

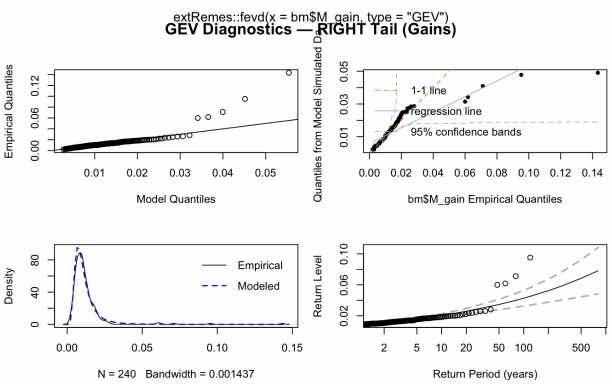


Figure 2.5. GEV Diagnostic Plots for the Right Tail (Extreme Gains)

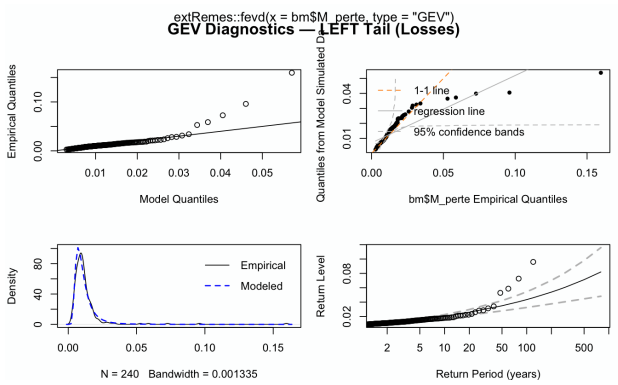


Figure 2.6. GEV Diagnostic Plots for the Left Tail (Extreme Losses)

However, while the GEV approach identifies the nature of the risk, it is inherently limited by the reduced sample size of monthly maxima. To gain greater statistical precision and leverage all available

extreme observations, we now turn to the Peaks Over Threshold (POT) method, which will allow us to refine these estimates and calculate operational risk measures such as Value-at-Risk (VaR).

2.3 Peaks Over Threshold (POT) Method

The Block Maxima approach, while robust, has a major limitation: it wastes a significant amount of information by retaining only a single extreme value per period (e.g., the worst monthly return), potentially ignoring other major shocks that occurred within the same timeframe.

To overcome this, we apply the Peaks Over Threshold (POT) method. As highlighted in the course (Chapter 1, page 16 of **coursetse2025-2026**), the POT method analyzes all observations that exceed a sufficiently high threshold u. According to the Pickands-Balkema-de Haan theorem (see), the distribution of these exceedances $(X - u \mid X > u)$ converges to a Generalized Pareto Distribution (GPD). This approach is particularly well-suited for analyzing exchange rates such as the USD/EUR, as it captures every volatility spike during periods of financial turbulence, thereby providing a much more granular view of extreme risk.

2.3.1 Selection of Threshold u

The selection of the threshold u is the most critical step in the POT method, as it relies on a classic bias-variance trade-off. A threshold that is too low includes non-extreme observations (introducing bias), while a threshold that is too high leaves too few observations for precise estimation (increasing variance).

To determine the optimal threshold, we combined two graphical diagnostic tools:

- **Mean Excess Plot (MEP):** If the exceedances follow a GPD, the expected value of these exceedances must be a linear function of the threshold. Graphically, we identified that the curve adopts a clear upward linear profile beyond the 95th percentile.
- **Parameter Stability Plot:** We plotted the evolution of the shape parameter ξ and the modified scale parameter against the number of upper order statistics (exceedances) k. (see)

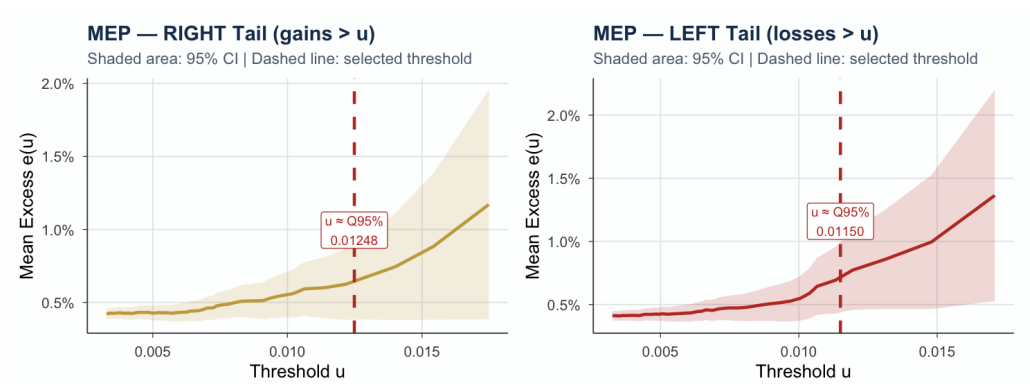


Figure 2.7. Mean Excess Plot.

The simultaneous analysis of these two graphs led us to select $k = 250$ exceedances for both tails (which roughly corresponds to the **95%** quantile of our total sample). This choice offers an optimal balance: the shape parameter ξ begins to stabilize around $k = 200$, and the scale parameter finds its stability around $k = 250$. We thus retain a sufficiently large sample, ensuring controlled variance for our estimations.

2.3.2 GPD Parameter Estimation

By applying the Maximum Likelihood Estimator (MLE) to the exceedances beyond our selected threshold ($k = 250$), we obtain the GPD parameters presented in Table 2.2.

Tail	**Threshold** u	**Exceedances** k	**Shape** ξ (SE)	**Scale** σ (SE)
Right (Gains)	0.0097	250	0.315 (0.071)	0.0035 (0.0003)
Left (Losses)	0.0096	250	0.209 (0.043)	0.0034 (0.0002)

Table 2.2. Maximum Likelihood Estimates of GPD parameters ($k = 250$).

Both estimated shape parameters ($\hat{\xi}$) are strictly positive, which confirms that both extreme gains and extreme losses fall within the Fréchet domain of attraction, indicative of heavy-tailed distributions. While the point estimate for the right tail ($\hat{\xi} \approx 0.315$) appears noticeably larger than that of the left tail ($\hat{\xi} \approx 0.209$), we cannot hastily conclude that the tails are asymmetric. At this stage, standard asymptotic procedures like the Wald test are unreliable due to the serial dependence present in our current daily data, which violates the underlying i.i.d. assumptions.

To rigorously address this issue, a formal symmetry test is deferred to Section 3. In that section, we will conduct the estimations and testing procedures using weekly average data—effectively filtering out the short-term dependence—allowing us to draw a definitive and statistically sound conclusion regarding the symmetry of the USD/EUR extreme risks.

2.3.3 Diagnostics and GPD Model Validation

To validate the goodness-of-fit of our GPD model to the empirical data, we examined standard diagnostic plots.

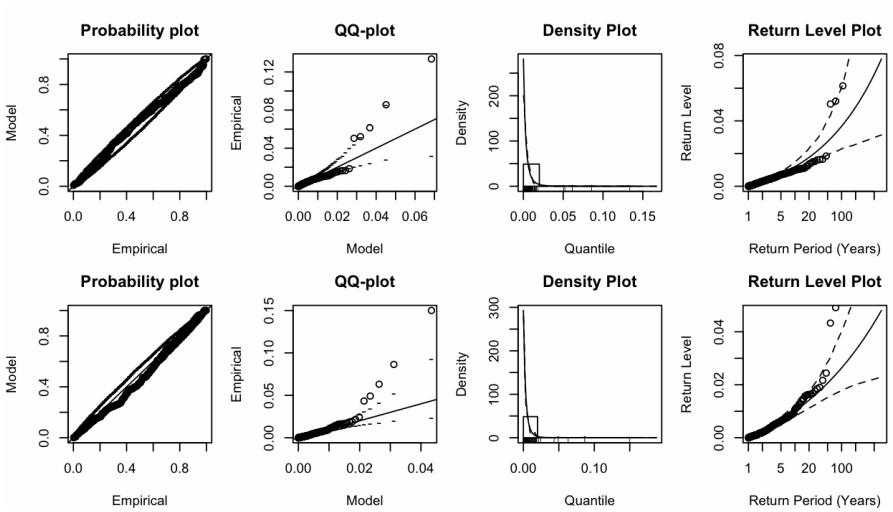


Figure 2.8. GPD diagnostic plots (in top for right tail and down for the left).

The overall fit is excellent. The vast majority of empirical quantiles align perfectly with the theoretical diagonal on the QQ-plots. We only observe a slight deviation for a handful of extreme points at the very end of the tail. This phenomenon is typical in financial Extreme Value Theory: these isolated points historically correspond to days of market crashes or major systemic crises. Outside of these singular extreme events, the GPD model accurately captures the tail dynamics.

2.3.4 Return Levels and Bootstrap Confidence Intervals

A crucial step in our study is translating these theoretical parameters into concrete risk metrics, notably through **Return Levels**. The T-year return level represents the quantile that is expected to be exceeded, on average, only once during a trading period of T years (assuming 252 trading days per year).

To compute the confidence intervals for these extreme quantiles, we opted for a bootstrap approach (with B=1000) on the exceedances. This methodological choice is supported by recent extreme value literature. Notably, de Haan et al. (2024) demonstrate that the bootstrap procedure yields statistically consistent confidence intervals within the Peaks-Over-Threshold (POT) framework.

Return	**Gains (USD Appreciation)**		**Losses (USD Depreciation)**	
Period	**Estimate**	**95% CI**	**Estimate**	**95% CI**
1 Year	2.34%	[1.84 ; 2.67]	2.08%	[1.85 ; 2.59]
2 Years	2.96%	[2.12 ; 3.62]	2.51%	[2.12 ; 3.65]
5 Years	4.03%	[2.58 ; 5.47]	3.20%	[2.47 ; 6.04]
10 Years	5.07%	[2.86 ; 7.48]	3.82%	[2.80 ; 9.17]
20 Years	6.39%	[3.19 ; 10.33]	4.55%	[3.06 ; 14.15]
50 Years	8.68%	[3.58 ; 16.38]	5.69%	[3.43 ; 25.23]
100 Years	10.95%	[3.89 ; 23.51]	6.72%	[3.70 ; 40.88]

Table 2.3. Estimated Return Levels and 95% Bootstrap CIs for USD/EUR daily returns.

Table 2.3 highlights a fundamental asymmetry in the USD/EUR exchange rate dynamics, particularly visible at the 20-year and 50-year horizons. A 20-year extreme event yields a 6.39% upward shock (USD appreciation), compared to a 4.55% downward shock. Extrapolating to a 50-year horizon, the gap widens significantly: a brutal 8.68% gain versus a 5.69% loss. Economically, this is consistent with the strong "Safe Haven" status of the US Dollar. During major systemic crises, massive flights-to-liquidity generate sudden and violent upward spikes in the USD value that far exceed the magnitude of its severe depreciations.

2.3.5 Extreme Risk Metrics: Value at Risk (VaR) and Expected Shortfall (ES)

To evaluate the implications of our results for monetary stability, we compute the Value at Risk (VaR) (see A.3) and Expected Shortfall (ES). These metrics are essential for central banks and financial institutions to monitor potential currency shocks that could destabilize the exchange rate regime.

Confidence	**Losses (Left Tail)**		**Gains (Right Tail)**	
Level	**VaR**	**ES**	**VaR**	**ES**
95.0%	0.93%	1.36%	0.96%	1.48%
99.0%	1.58%	2.19%	1.70%	2.57%
99.5%	1.94%	2.65%	2.16%	3.25%
99.9%	3.02%	4.02%	3.73%	5.56%

Table 2.4. Daily VaR and Expected Shortfall (ES) based on the fitted GPD models.

Table 2.4 presents the VaR and Expected Shortfall (ES) estimates for different confidence levels. ES is consistently higher than VaR, a difference that widens significantly at 99.9% (increasing from 3.73% to 5.56% for gains). This difference indicates that extreme shocks to the USD/EUR pair are not only rare but also of a much higher average magnitude once the risk threshold is crossed.

However, these marginal risk measures should be interpreted with caution at this stage. While the quantile calculation remains theoretically valid, the presence of serial dependence in the daily data may affect the stability of these estimates.

3 Handling Serial Dependence

Financial time series, particularly exchange rates, exhibit significant serial dependence and volatility clustering. Such features violate the i.i.d. assumption crucial for Extreme Value Theory (EVT). This temporal dependence makes classical confidence intervals unreliable. To obtain valid intervals and conduct robust statistical tests, we implement two aggregation strategies: quarterly block maxima and weekly average returns. These approaches remove serial memory and ensure reliable inference.

3.1 Data aggregation approaches and Validation of Independence

we established two distinct temporal aggregation approaches aimed at restoring the i.i.d. assumption: selecting the maximum return within each 3-month period (**Quarterly Block Maxima**) and aggregating data into weekly average returns (**Weekly POT**). To validate the robustness of these strategies, we conducted Box-Pierce tests on both the raw filtered data (to check for linear autocorrelation) and their squares (to detect residual conditional heteroskedasticity, or ARCH effects).

Data aggregation Method	P-value (Raw)	P-value (Squared)
Block Maxima (Quarterly)	0.9525	0.9999
Weekly Returns (POT)	0.2880	0.0000

Table 3.1. Box-Pierce test results (lag=30) for independence validation.

As shown in Table 3.1, the Quarterly Block Maxima approach effectively removes both linear dependence and volatility clustering, failing to reject the null hypothesis of independence for both raw ($p = 0.9525$) and squared returns ($p = 0.9999$). Conversely, while the Weekly POT method eliminates linear autocorrelation ($p = 0.2880$), persistent ARCH effects remain in the squared returns ($p = 0.0000$).

Navigating between these frameworks illustrates the classic bias-variance tradeoff inherent to data aggregation: shifting from monthly to quarterly Block Maxima slashes our sample from 240 to merely 80 observations (minimizing bias but maximizing estimator variance), whereas the weekly POT reduces 5,000 daily points to roughly 1,040 observations to strike a practical balance. Despite this necessary data loss, successfully mitigating temporal dependence ensures the validity of asymptotic properties. Consequently, we can reliably construct asymptotic confidence intervals or, more crucially for our analysis, perform formal statistical inferences such as the Wald test to test tail symmetry ($H_0 : \xi_{right} = \xi_{left}$).

3.2 Statistical Inference: Tail Heaviness and Symmetry

Having established the statistical validity of our declustering frameworks (Quarterly BM and Weekly POT), we can now rely on the asymptotic properties of our estimators to perform formal inferences. We test two fundamental properties of the USD/EUR returns: the "heavy" nature of the distribution tails ($H_0 : \xi = 0$) and their bilateral symmetry ($H_0 : \xi_{right} = \xi_{left}$).

The diagnostic plots for both the Quarterly BM and Weekly POT frameworks (detailed in Figures B.3 and B.5) yield highly satisfactory results; the empirical data closely follow the theoretical distributions, with only a few minor deviations at the extreme ends of the tails, thereby validating the robustness of our models for further statistical inference.

	Right Tail (Extreme Gains)		**Left Tail (Extreme Losses)**		**Symmetry Test**
Method	$\hat{\xi}_{right}$	p-value ($\xi = 0$)	$\hat{\xi}_{left}$	p-value ($\xi = 0$)	Wald p-value
BM Quarterly (GEV)	0.3702	< 0.001	0.3000	< 0.001	0.5832
Weekly POT (GPD)	0.2358	0.0658	0.3160	0.0238	0.7196

Table 3.2. Statistical inference on tail indices: Testing for tail heaviness ($H_0 : \xi = 0$ vs $H_1 : \xi > 0$) and tail symmetry ($H_0 : \xi_{right} = \xi_{left}$ vs $H_1 : \xi_{right} \neq \xi_{left}$).

1. Test of Tail Heaviness (Fréchet Domain)

While the heavy-tailed nature of financial returns is very often taken for granted (in all our analysis, we got a EVI larger than 1) , formally testing $H_0 : \xi = 0$ confirms their alignment with the Fréchet domain. Under the strictly independent Quarterly BM framework, the p-values are strictly below 0.001, firmly rejecting the hypothesis of an exponential decay (Gumbel domain).

Interestingly, the Weekly POT approach strongly confirms this heaviness for the left tail ($p = 0.0238$), but is only significant at the 10% level for the right tail ($p = 0.0658$). Rather than challenging the Fréchet assumption (the point estimate $\hat{\xi} = 0.2358$ remains substantially positive), this perfectly illustrates the expected loss of statistical power due to data aggregation. As established in our Box-Pierce diagnostics, ARCH effects (conditional variance) persist in the weekly series. This dependence, combined with a severely restricted number of exceedances ($k = 52$), inflates the estimator's standard error.

2. Test of Tail Symmetry

Although the $\hat{\xi}$ point estimates in Table 3.2 display slight numerical differences between extreme gains and losses, the Wald test formalizes this comparison by accounting for estimation uncertainty. The resulting p-values (0.5832 for BM and 0.7196 for POT) clearly indicate a failure to reject the null hypothesis of symmetry at the 5% level.

Conclusion

This study aimed to characterize the extreme behavior of the USD/EUR exchange rate over the period 2004–2023 using Extreme Value Theory (EVT). The analysis highlights several key findings regarding the statistical properties and risk dynamics of foreign exchange returns.

First, the exploratory analysis shows that the empirical distribution of log-returns clearly deviates from normality. Graphical diagnostics and formal tests confirm excess kurtosis and tails that are thicker than those of a normal distribution. This finding invalidates standard Gaussian-based risk models and justifies the use of Extreme Value Theory as a more appropriate framework for modeling extreme currency fluctuations.

Second, both right and left tails of the distribution belong to the Fréchet domain of attraction. Across all estimation methods (Hill, GEV, and GPD), the tail index ξ is consistently positive. This confirms that extreme exchange rate movements follow a heavy-tailed, power-law behavior, implying that large shocks are significantly more likely than predicted by light-tailed models.

Third, the economic magnitude of extreme events is substantial. Return level estimates suggest that daily exchange rate variations exceeding 5% are plausible over a 10-year horizon. This finding is reinforced by the daily Value-at-Risk (VaR) and Expected Shortfall (ES): at the 99% confidence level, the VaR reaches 1.58% for losses and 1.70% for gains, while the ES rises to 2.19% and 2.57%, respectively. These results underscore the importance of accounting for tail risk in financial risk management, particularly for institutions exposed to foreign exchange markets.

Finally, addressing serial dependence proves essential for valid inference. Financial returns exhibit volatility clustering and autocorrelation, violating the i.i.d. assumption underlying EVT. By implementing data aggregation techniques (Block Maxima quarterly and weekly average returns POT), we obtain more reliable estimators and enable formal statistical testing. Although point estimates of the tail index differ slightly between gains and losses, formal statistical tests fail to reject the null hypothesis of equality. This suggests that extreme appreciations and depreciations of the USD/EUR exchange rate exhibit comparable tail behavior, consistent with the high liquidity and efficiency of major currency markets.

Despite these results, several limitations remain. Data aggregation reduces the effective sample size and may increase estimation variance. Moreover, residual heteroskedasticity suggests that a *filtered EVT* approach (combining GARCH models with EVT) could further improve the accuracy of tail estimation. Future research could also extend the analysis to non-stationary EVT models incorporating macroeconomic variables or regime-switching dynamics.

In conclusion, the USD/EUR exchange rate exhibits clear heavy-tailed behavior with significant extreme risk on both sides of the distribution. EVT provides a powerful and robust framework to quantify these risks, offering valuable insights for financial regulation, portfolio management, and currency risk hedging strategies.

References

[1] Bank for International Settlements. (2022). *Triennial Central Bank Survey: OTC Foreign Exchange Turnover in April 2022*. Statistical Release, BIS, Basel. https://www.bis.org/statistics/rpfx22_fx.htm

[2] Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer Series in Statistics. Springer-Verlag London.

[3] de Haan, L., & Zhou, C. (2024). Bootstrapping Extreme Value Estimators. *Journal of the American Statistical Association*, 119(545), 382–393. https://doi.org/10.1080/01621459.2022.2120400

[4] Manganelli, S., & Engle, R. F. (2001). Value at Risk models in finance. *European Central Bank Working Paper Series*, No. 75. European Central Bank. https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp075.pdf

[5] Beirlant, J., Goegebeur, Y., Segers, J., & Teugels, J. (2004). *Statistics of Extremes: Theory and Applications*. Wiley Series in Probability and Statistics.

[6] Walguen, O. (2021). *Analyse statistique des valeurs extrêmes de processus spatio-temporels* [Thèse de doctorat, Université des Antilles]. Archive ouverte HAL. https://theses.hal.science/tel-05005412v1/file/OSCAR_2021.pdf

[7] Delcaillau, D., & Tioguim, I. (2018). *Modélisation d'extrêmes de séries temporelles : une étude empirique*. Bureau d'études EURIA, Université de Bretagne Occidentale / IFREMER. https://www.univ-brest.fr/euria/sites/euria.www.univ-brest.fr/files/2022-06/etude_empririque_2017-2018.pdf

[8] Girard, S. (2011). *Introduction à la statistique des valeurs extrêmes* [Support de présentation]. Équipe-projet Mistis, INRIA Rhône-Alpes & Laboratoire Jean Kuntzmann (LJK). https://mistis.inrialpes.fr/docs/slidesGrenoble.pdf

[9] Daouia, A. (2025–2026). *Extreme Value Analysis: Course and practical sessions*. M2 Econometrics Statistics, Toulouse School of Economics.

A Theoretical Appendix: Foundations of Extreme Value Theory

Extreme Value Theory (EVT) characterizes the asymptotic behavior of extreme realizations in a sample, relying on two fundamental theorems.

A.1 Fisher-Tippett-Gnedenko Theorem (Limit of Maxima)

Theorem 1 — Convergence of Normalized Maxima

Let $(X_i)_{i\geq 1}$ be an *i.i.d.* sequence with distribution F, and $M_n = \max(X_1, \ldots, X_n)$. If there exist sequences $a_n > 0$ and $b_n \in \mathbb{R}$ such that:

$$\frac{M_n - b_n}{a_n} \xrightarrow{d} G, \qquad n \to \infty, \tag{A.1}$$

where G is a non-degenerate distribution, then G must belong to the *Generalized Extreme Value* (GEV) family:

$$\boxed{G_{\mu,\sigma,\xi}(x) \;=\; \exp\left\{-\left[1+\xi\,\frac{x-\mu}{\sigma}\right]_+^{-1/\xi}\right\}} \tag{A.2}$$

with location $\mu \in \mathbb{R}$, scale $\sigma > 0$, and shape $\xi \in \mathbb{R}$ ($[z]_+ = \max(z, 0)$).

The shape parameter ξ governs the tail behavior, classifying distributions into three max-domains of attraction:

- **Fréchet ($\xi > 0$):** Heavy, power-law tails (e.g., Pareto, Student-t).
- **Gumbel ($\xi = 0$):** Light, exponentially decaying tails (e.g., Normal).
- **Weibull ($\xi < 0$):** Bounded finite upper tails (e.g., Uniform).

A.2 Pickands-Balkema-de Haan Theorem (Limit of Exceedances)

Theorem 2 — Threshold Exceedances Distribution

If F belongs to the max-domain of attraction of a GEV(ξ), then for a sufficiently high threshold u, the conditional excess distribution $Y = X - u \mid X > u$ converges to the *Generalized Pareto Distribution* (GPD):

$$\boxed{H_{\sigma_u,\xi}(y) \;=\; 1-\left(1+\xi\,\frac{y}{\sigma_u}\right)_+^{-1/\xi}, \quad y \geq 0} \tag{A.3}$$

with $\sigma_u = \sigma + \xi(u - \mu) > 0$.

Remark: The shape parameter ξ strictly mirrors the GEV parameter, creating a direct theoretical bridge between block maxima and threshold exceedances models.

A.3 Relevance for Financial Risk Management

Standard normal assumptions systematically underestimate extreme financial shocks. EVT provides robust, asymptotically unbiased estimators for extreme quantiles like the Value at Risk (VaR), which is crucial for regulatory compliance at high confidence levels $(1 - \alpha)$:

$$\underbrace{\mathrm{VaR}_\alpha^{\mathrm{GPD}}}_{\mathrm{EVT}} = u + \frac{\hat{\sigma}}{\hat{\xi}}\left[\left(\frac{1-\alpha}{\hat{p}_u}\right)^{-\hat{\xi}} - 1\right] \quad \gg \quad \underbrace{z_\alpha \cdot \hat{\sigma}_r + \hat{\mu}_r}_{\text{Gaussian Approximation}} \tag{A.4}$$

B Supplementary Material Appendix

Test	Statistic	p-value	Conclusion
Augmented Dickey-Fuller (ADF)	-16.8164	< 0.01	Stationary ✓
Phillips-Perron (PP)	-5810.9681	< 0.01	Stationary ✓
KPSS	0.0663	> 0.10	Stationary ✓

Table B.1. Unit root and stationarity tests on log-returns (r_t)

Test (Lag = 20)	Statistic	p-value	Conclusion
Ljung-Box on r_t	$Q = 220.50$	< 0.01	Autocorrelation
Ljung-Box on r_t^2	$Q = 1246.34$	< 0.01	ARCH effects detected
ARCH-LM test on r_t	$\chi^2 = 1895.40$	< 0.01	ARCH effects detected

Table B.2. Dependence and heteroskedasticity tests

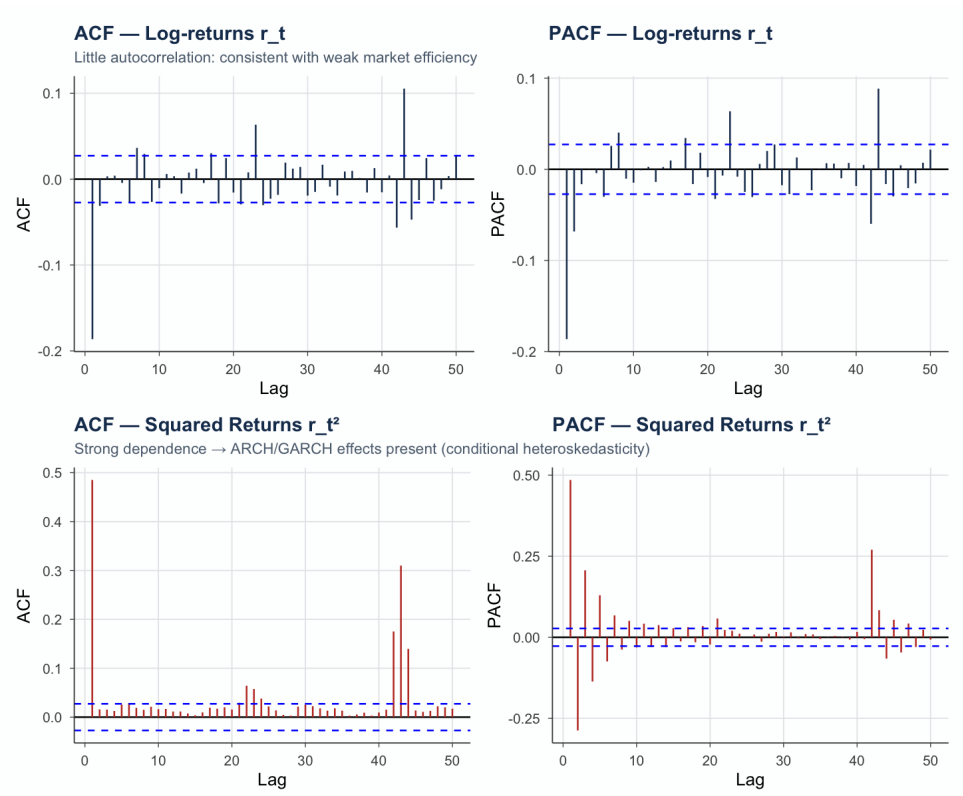


Figure B.1. ACF and PACH of returns and squared returns

Test	Statistic	p-value	Conclusion
Jarque-Bera	2 070 874.27	< 0.001	Rejection of normality (H_0)
Kolmogorov-Smirnov	0.0847	< 0.001	Rejection of normality (H_0)
Anderson-Darling	107.70	< 0.001	Rejection of normality (H_0)
Shapiro-Wilk*	0.7758	< 0.001	Rejection of normality (H_0)

** Shapiro-Wilk test performed on a random subsample of* 5000 *observations.*

Table B.3. Formal normality tests on USD/EUR daily log-returns

Statistic	Value	Normal Baseline	Indication
Skewness	-0.5851	0	Left-skewed (Asymmetric)
Kurtosis (raw)	101.0217	3	–
Excess Kurtosis	98.0217	0	Heavy tails suspected

Table B.4. Shape statistics of USD/EUR daily log-returns

Method	Right Tail (Gains)	Left Tail (Losses)
Hill (k^*)	0.3093	0.3542
RB-Hill (k^*)	0.2868	0.3292
Moment DEdH (k^*)	0.4626	0.4767
dAMSE (optimal)	0.3093	0.3542

Table B.5. Summary of tail index (γ) estimators at optimal threshold k^*

Table B.6. Asymptotic Wald Test for Tail Symmetry ($H_0 : \xi_{\text{Right}} = \xi_{\text{Left}}$)

Block Size	$\hat{\xi}_{\text{Right}}$	$\hat{\xi}_{\text{Left}}$	Difference	Z-stat	p-value
Monthly (n=240)	0.2576	0.2799	-0.0223	0.321	0.748
Quarterly (n=80)	0.3702	0.3000	0.0702	0.549	0.583

Note: The Wald test is based on the asymptotic normality of the Maximum Likelihood estimators. A p-value > 0.05 *indicates that we fail to reject the null hypothesis of symmetry at the 5% confidence level. Despite the numerical increase of* ξ *with quarterly blocks, the statistical conclusion remains robust.*

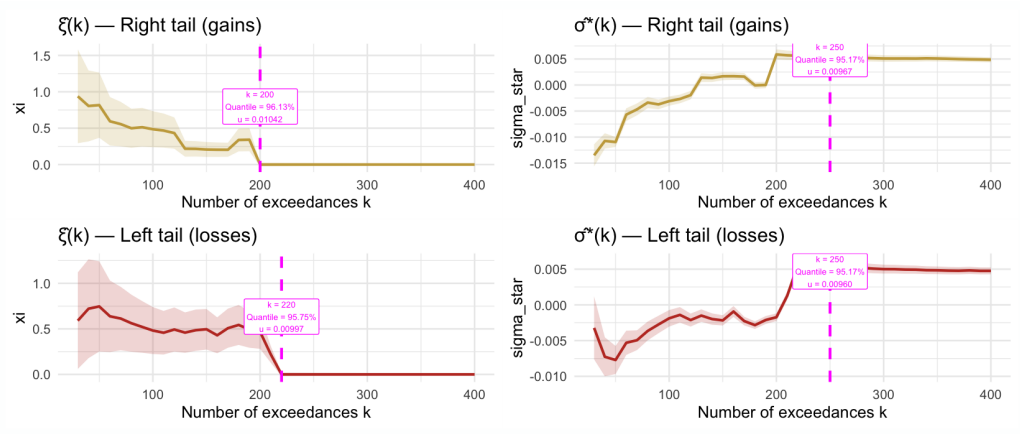
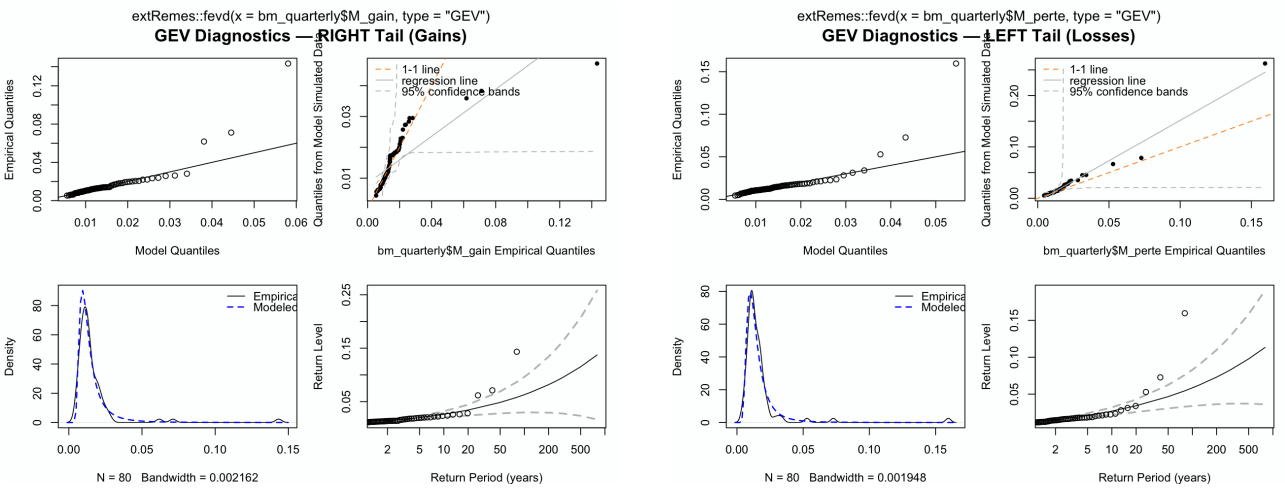


Figure B.2. Mean Excess Plot and GPD Parameter Stability Plot.



(a) Right Tail (Extreme Gains) **(b)** Left Tail (Extreme Losses)

Figure B.3. GEV Diagnostic Plots: Quarterly Block Maxima framework. The alignment of empirical points with the theoretical diagonal in the QQ-plots validates the Fréchet model fit.

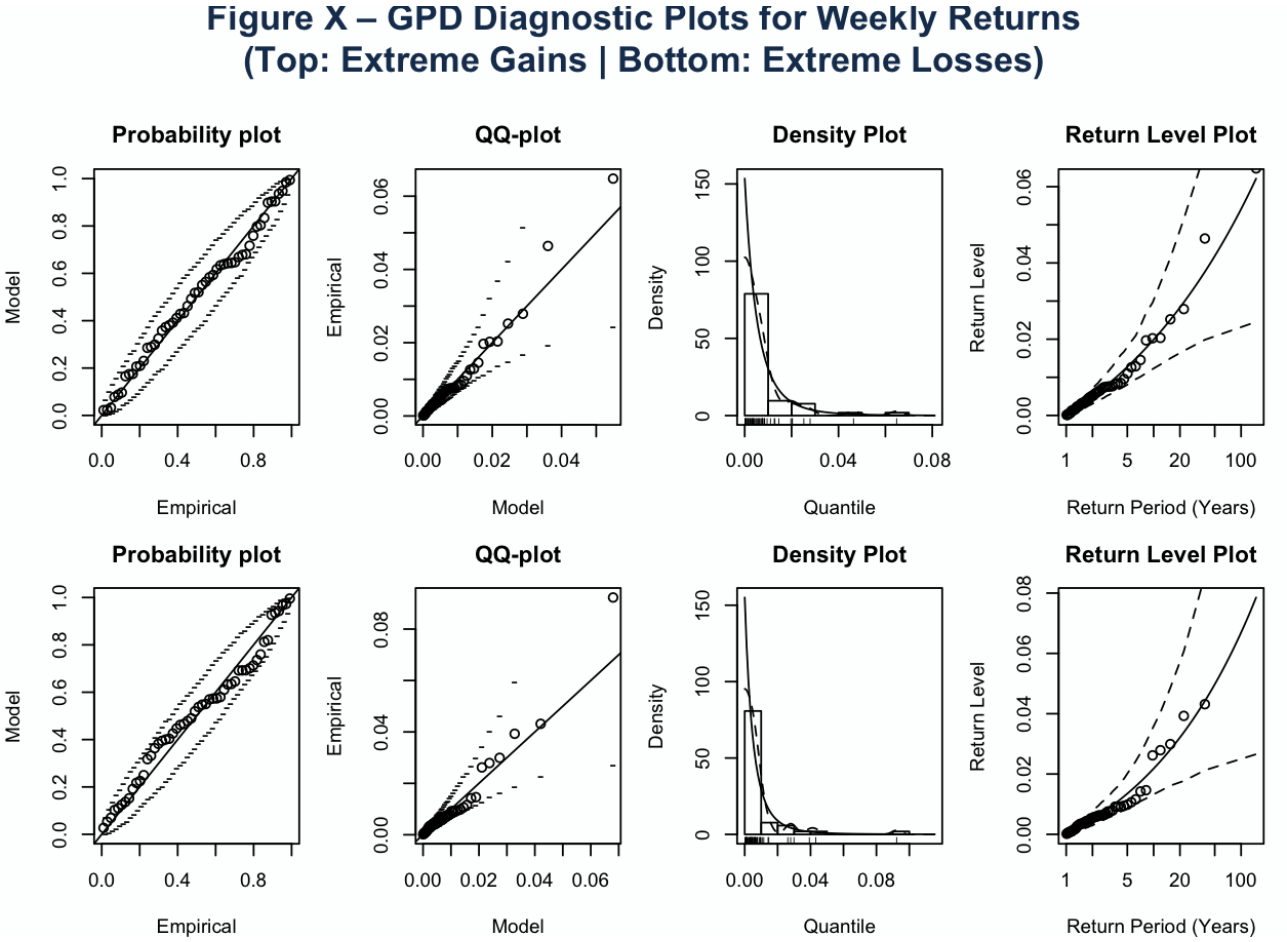


Figure B.4. Right Tail (Extreme Gains)

Figure B.5. GPD Diagnostic Plots: Weekly Peaks-Over-Threshold framework. Despite the lower sample size, the return level plots remain within the 95% confidence intervals.